SUBJECT: financial management 3

DATE: / /

1) liquidity ratios:-

- $CR = \frac{CA}{CL}$
- Quick acid test: $\frac{CA - INV}{CL}$
- cash ratio $= \frac{cash}{CL}$

2) Asset managment:-

- Inventory turnover $= \frac{CGS}{INV}$
- T.A turnover $= \frac{Sales\ Rev}{TA}$
- Fixed asset turnover $= \frac{SR}{NI + FA}$
- DSO $= \frac{AR}{\frac{credit\ sales}{360}}$

3) debt managment ratios:-

The ability of the business to meet its debt obligation

interest + principal

"serving debt"

① debt ratio

$= \frac{total\ debt\ (TD)}{Total\ asset}$

"better to be low" → default risk

Note: $TA = CA + NFA$

$TA = TL + OE$

financial leverage

Note: capital intensive industries → high debt ratios

$\frac{TD}{TA} = \frac{70}{?}$

if Averag = 70

- 80 → worse off
- 60 → better off

0,70 TA = debt

0,30 TA = OE

* $\frac{equity}{TA} = 1 - \frac{debt}{TA}$

* equity multiblier (Em) $= \frac{TA}{equity} = 1 + \frac{debt}{equity}$

ELZAHRAA

(x) debt ratio $= \frac{TD}{TA} = 0,55$, $\frac{debt}{equity} = ?$, equdity multiplier

$\frac{TD}{TA} = 0,55$

$0,55\,TA = debt$

$0,45\,TA = equity$

$$\frac{debt}{equit} = \frac{0,55\,TA}{0,45\,TA} = 1,22$$

$$equity\ multiplier = \frac{TA}{equity} = 1 + \frac{debt}{equity} = 2.22$$

② Time interest earn (TIE) / interest coverag:

$$= \frac{\text{earn before interest and TAX (EBIT)}}{\text{Interest coverage}}$$

Sales Rev	XX
- CoGS	(XX)
Gross profit	XX
- all exp before dep, interest, TAX	(XX)
EBDIT	XX
- Dep exp	(XX)
EBIT	XX
- interest	(XX)
EBT	XX
- tax exp	(XX)
Net income	XX

Note: EBDIT because it's non cash

Note: net income = dividends + Retained earnings

③ Cash coverage

$$\frac{\text{EBDIT}}{\text{interest coverage}}$$

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ex Sales = 14000000 , TA = 6000000 , TA turnover ?

$$\text{TA turnover} = \frac{\text{Sales Rev}}{\text{TA}} = 2,33 \text{ x}$$

عايزة تخليه 2.75x وتسيب الـ 6M زي ماهي

$$\frac{\text{Sales}}{\text{TA}} = \frac{\text{Sales}}{6000000} = 2.75$$

$$\text{Sales} = 16500000$$

ex TA = 6M , $\frac{CA}{Cl} = 1.5x$, quick ratio = 1

$$Cl = 4M$$

$$\frac{CA - INV}{Cl} = 1$$

$$\frac{6M + INV}{4M} = 1$$

$$INV = 2M$$

ex $\frac{\text{Sales}}{TA} = 2.5x$, TL + OE = 1.5

TA = 1.5

$$\frac{\text{sales}}{TA} = 2.5$$

$$\frac{\text{Sales}}{1.5M} = 2.5 \quad \therefore \text{Sales} = 3750000$$

3) profitability ratios

- *profit margin
  - $\frac{NI}{\text{Sales Rev}} = -\%$
- *Return on asset RoA
  - $\frac{NI}{TA} = -\%$
- *Return on Equity RoE
  - $\frac{NI}{\text{equity}} = -\%$

(ex) if the $\frac{CA}{Cl} = 1.2\ \text{times}$, Cl = 20000, industry Average = 1.9

$\frac{CA}{20000} = 1.2$ ∴ CA = 24000 ("worse off")

if CA: 24000 < Cash 4000, AR 20000 — Quick Ratio

Quick Ratio = $\frac{CA - INV}{Cl}$ → 1000 → CA − cash − AR / cash + AR

(ex) Net working capital (Nwc) = 850 (CA − Cl), Cl = 4700, CA ratio?

850 = CA = 4700x

CA = 5550

CA ratio = $\frac{CA}{Cl} = \frac{5550}{4700} = 1.18$ x average = 1.9 better off

Quick ratio? INV = 1600

= $\frac{CA - INV}{Cl} = \frac{5550 - 1600}{4700}$